



Myelin water imaging data analysis in less than one minute

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ABSTRACT

Purpose: Based on a deep learning neural network (NN) algorithm, a super fast and easy to implement data analysis method was proposed for myelin water imaging (MWI) to calculate the myelin water fraction (MWF).

Methods: A NN was constructed and trained on MWI data acquired by a 32-echo 3D gradient and spin echo (GRASE) sequence. Ground truth labels were created by regularized non-negative least squares (NNLS) with stimulated echo corrections. Voxel-wise GRASE data from 5 brains (4 healthy, 1 multiple sclerosis (MS)) were used for NN training. The trained NN was tested on 2 healthy brains, 1 MS brain with segmented lesions, 1 healthy spinal cord, and 1 healthy brain acquired from a different scanner.

Results: Production of whole brain MWF maps in approximately 33 s can be achieved by a trained NN without graphics card acceleration. For all testing regions, no visual differences between NN and NNLS MWF maps were observed, and no obvious regional biases were found. Quantitatively, all voxels exhibited excellent agreement between NN and NNLS (all $R^2 > 0.98$, $p < 0.001$, mean absolute error < 0.01).

Conclusion: The time for accurate MWF calculation can be dramatically reduced to less than 1 min by the proposed NN, addressing one of the barriers facing future clinical feasibility of MWI.

1. Introduction

Myelin water, the water trapped in myelin bilayers, can be used as a myelin biomarker (Whittall and MacKay, 1989). Myelin water imaging (MWI) (Whittall and MacKay, 1989; MacKay et al., 1994) is a quantitative magnetic resonance (MR) method that measures myelin water and has been used extensively to study many myelin-related diseases in both brain and spinal cord, including multiple sclerosis (MS) (Vargas et al., 2015; Laule et al., 2010), neuromyelitis optica spectrum disorder (Manogaran et al., 2016), schizophrenia (Lang et al., 2014), phenylketonuria (Sirrs et al., 2007), Niemann-Pick disease (Davies-Thompson et al., 2016), Krabbe disease (Laule et al., 2018), neurofibromatosis-1 (Billiet et al., 2014), progressive solitary sclerosis (Lee et al., 2019), stroke (Borich et al., 2013), Parkinson's disease (Baumeister et al., 2018), concussion (Wright et al., 2016), and cervical spondylotic myelopathy

(Liu et al., 2017). Typically, MWI acquires multi-echo data to characterize the multi-exponential T_2 decay signals generated from different water compartments within each imaging voxel. Image post-processing is then performed to separate the signal of the myelin water compartment, which exhibits shorter T_2 relaxation times from that of the intra- and extra- (IE) cellular water, from the total water signal. Over the years, various sequences have been used in MWI, including Carr-Purcell-Meiboom-Gill (CPMG) (Whittall and MacKay, 1989), 3D gradient and spin echo (GRASE) (Prasloski et al., 2012a), T_2 preparation (T_2 prep) (Oh et al., 2006), and multi-gradient echo (MGRE) (Hwang et al., 2010). For MWI data analysis, many fitting algorithms (Whittall and MacKay, 1989; Raj et al., 2014; Akhond-Asl et al., 2014; Nataraj et al., 2018a; Du et al., 2007) have been developed to decompose the voxel-wise multi-exponential decay data into a spectrum of T_2 components, from which the myelin water fraction (MWF, the ratio of myelin

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water signal to total signal), is of the most interest and can be extracted.

Non-negative least squares (NNLS) T_2 fitting algorithm (Lawson and Hanson, 1995; Provencher, 1982; Kroeker and Mark Henkelman, 1986), first introduced to MWI by Whittall and Mackay (Whittall and MacKay, 1989), is a commonly used analysis method. NNLS makes no prior assumptions about the total number of distinguishable water compartments and their corresponding T_2 values, which is a highly desirable feature especially when studying a complex biological system such as pathological lesions with heterogeneous composition. However, having very few initial assumptions leads to an underdetermined problem with non-unique solutions, making the NNLS MWF calculation results unstable. Alternatively, many other fitting algorithms have subsequently been developed such as the spatially constrained multi-Gaussian model by Raj et al. (2014), the Wald distribution by Akhondi-Asl et al. (2014), and the quasi-Newton algorithm by Du et al. (2007). These methods usually offer a more stable MWF estimation but at the expense of presuming only two or three water compartments (myelin water, IE water and free water) as a necessary initial assumption. The stability of NNLS was improved by using regularization to constrain the data misfit (Tikhonov, 1963; Whittall et al., 1997), and further improvements have been made in recent years by applying spatial regularization (Hwang and Du, 2009) as well as non-local spatial regularization (Yoo and Tam, 2013). The latest NNLS analysis also incorporates the extended phase graph algorithm to estimate the refocusing flip angle of each voxel to correct for the effect of stimulated echoes (Prasloski et al., 2012b). Overall, there is no single perfect fitting algorithm to the ill-posed T_2 decomposition problem and each method has its own strengths and weaknesses. For this proof of concept paper, we focused on the NNLS-based analysis method due to its high popularity (Liu et al., 2019; Liu et al., 2020) and good reproducibility (Lee et al., 2018), as well as the fact of it being histologically validated (Laule et al., 2006, 2008), and used T_2 data from a common acquisition approach, the 3D GRASE sequence (Prasloski et al., 2012a).

Despite reductions in the whole brain scan time of 3D GRASE to under 8 min (Zhang et al., 2015), clinical use of MWI for MS and other demyelinating diseases has not yet been possible for a number of reasons, one of which is related to the speed and complexity of MWI data analysis. Fitting noisy MR data with a simple NNLS algorithm is typically not good enough because NNLS is very susceptible to noise, leading to unstable MWF results and unsmooth appearance of the resulting MWF map. Regularized NNLS fitting with stimulated echo correction (Prasloski et al., 2012b) is, for the most part, required by today's standards. But the mathematical complexity of such integration makes it very difficult to implement for new users. More importantly, the computational time required for whole brain data analysis using NNLS typically takes several hours for a modern workstation PC, which is unacceptable in clinical settings. Recent effort made by Yoo et al. (2015) using multicore CPU and graphics card parallelization can accelerate the NNLS analysis by approximately 4 times, and produce similar MWF values with small and explainable differences. However, a successful implementation of this method requires high-end hardware integration and relevant expertise in computing, which is also not always clinically feasible.

Another approach to shorten MWF analysis time is to use a neural network (NN) (Schmidhuber, 2015) algorithm, which is a supervised deep learning method that has attracted much attention in various fields including MR. A typical NN consists of an input layer to take the numerical input data, an output layer to yield the NN calculation results, and in between, there are hidden layers which are optimized by labelled training data to acquire the ability of mapping from the input data space to the desired output space. The basic building blocks for each layer are called nodes, which are connected between adjacent layers. The strength of the connection is governed by weights and biases which need to be optimized during training. The number of nodes and layers in a NN can be varied depending on the complexity of the problem. Before training, the activation function of each layer is defined to produce the output of each layer and introduce nonlinear properties to the NN. The loss function is defined to measure the error between the NN predicted values and

the ground truth labels. During training, an iterative process is taken to minimize the loss function, and the step size of each iteration towards the global minimum of the loss function is determined by the learning rate. The data of a typical deep-learning study is partitioned into 3 groups, namely the training set, validation set, and testing set. The training set is used during the iterative training process of the NN. If the training set has a very large size and is fed into the NN all at once, the training may become difficult and time-consuming. To solve this problem, the training set can be split into many data blocks (also known as mini batches), each of which usually contains 128, 256 or 512 training examples. Training on the mini batches is a commonly used approach when it comes to large-sized training sets. The validation set is used during training, to assess the prediction accuracy of the NN during each training iteration; during validation the loss, or error rate, is determined at each step which prevents overfitting. The training continues until the accuracy on validation data stops improving. Finally, the testing set is only used once, after NN training is completed. Such data partitioning ensures the testing data is obscured from the whole training process so that the one-time testing result is a true measurement of the actual predictive power of the trained NN. For additional background on machine learning and NN the reader is pointed to several comprehensive references (Schmidhuber, 2015; Lee et al., 2017).

In recently years, NN has been successfully implemented in many MRI applications from image acquisition to data analysis (Lundervold and Lundervold, 2019), most of which outperformed conventional methods in terms of faster speed and better accuracy. In our study, we proposed to train a deep learning NN to calculate MWF to overcome the aforementioned limitations of MWI. We will use the latest NNLS algorithm (Prasloski et al., 2012b) to produce the ground truth labelled training data, so that the trained NN can (1) produce faster voxel-wise MWF calculations on new data and (2) be easier to implement with common computer compatibility. In this paper, we examined the accuracy of the proposed NN as a faster and computationally compact representation of the commonly used NNLS method (Hornik et al., 1989). We also explored the feasibility of applying our NN trained by healthy brain data to calculate MWF for other central nervous system tissues including MS lesions and spinal cord, as well as data collected from a different scanner.

2. Methods

2.1. MWI data acquisition and pre-processing

2.1.1. Training data

MRI data for NN training were collected at 3T using an 8-channel SENSE head coil (Philips Achieva), including 4 healthy volunteers (4 males, age range 29–50 years) and 1 MS volunteer (female, age 39). MWI was performed using a 3D GRASE sequence (32 echoes, $TE1/\Delta TE/TR = 10/10/1000$ ms, axial matrix size = 232×186 , acquired resolution = $1 \times 1 \times 5$ mm³ for 20 slices, reconstructed resolution = $1 \times 1 \times 2.5$ mm³ for 40 slices, EPI factor 3, slice oversampling factor 1.3, SENSE factor 2, acquisition time = 14.4 min) (Prasloski et al., 2012a). The five brains were extracted by using the 'bet' tool in FSL software (Smith, 2002) on the first echo image to remove the skull and extract the brain tissue. This mask was then propagated to the rest of the echoes in the train. 32-echo decay data from all voxels in each brain were subsequently extracted using in-house software created with Python. In total, there were 2,952, 684 voxels for training.

2.1.2. Testing data

To illustrate the generality of the trained NN, we collected testing data from two additional healthy brains (2 males, 55 and 68 years old, total 1,182,365 voxels), one brain with MS lesions (female, 26 years old, lesion regions: 7838 voxels), and one healthy cervical spinal cord (male, 29 years old, C2 to C3 segment: 1433 voxels) from the same scanner (Philips Achieva). The testing brain data was acquired using the same 3D GRASE sequence and the same data extraction steps described above. The

spinal cord MWI data were acquired with a modified 3D GRASE sequence (32-echo, TE1/ΔTE/TR = 10/10/1500 ms, 8 axial slices, FOV = 180 mm × 150 mm × 40 mm, acquired resolution = 0.75 × 0.75 × 5 mm³, reconstructed resolution = 0.63 × 0.63 × 5 mm³, slice oversampling factor 1.3, SENSE factor 2, acquisition time = 8.5 min) (Ljungberg et al., 2017). The MS lesions were first manually segmented on the proton density image of the testing MS brain, and the spinal cord was segmented using Spinal Cord Toolbox (De Leener et al., 2017). Segmented tissue was then transformed into the GRASE space via rigid registration. Voxel-wise 32-echo decay data within the registered lesion mask (7838 voxels) and spinal cord mask (1433 voxels) were subsequently extracted.

2.1.3. Multi-site data

To demonstrate the feasibility of multi-site usage of the trained NN, MWI brain data from another healthy volunteer (female, 31 years old, 528,229 voxels) were collected using a different MRI scanner. Ideally, an MRI scanner manufactured by a different vendor would be best for illustrating this point. However, we only had access to another MR system from the same manufacturer but with a different generation (3T Philips Ingenia Elition X) and a different receive coil (32-channel SENSE head coil) to illustrate the concept of multi-site usage. The identical brain GRASE sequence and data processing were used as described above.

2.2. Ground truth

Voxel-wise 32-echo decay data were analyzed by regularized NNLS (MatLab) with stimulated echo correction (Prasloski et al., 2012b) (analysis program can be request from this URL: <https://mrresearch.mcgill.ca/news-projects/myelin-water-fraction/>) to compute MWF as the ground truth label for each voxel. One should be aware the MWF value generated by NNLS is just an estimation. It is the established convention in the deep learning field for training labels to be called the ‘ground truth’, and so we will refer to the NNLS calculated MWF values as ground truth as these are the values the NN is being trained to replicate. Using the phrase ‘ground truth’ is only because NNLS was our reference method, which a trained NN was trying to replicate. The NNLS algorithm works in a way that the misfit of χ^2 (MacKay et al., 1994) is minimized by the following equation:

$$\chi^2 = \sum_{i=1}^N \left| \sum_{j=1}^M A_{ij} s_j - y_i \right|^2 + \mu \sum_{j=1}^N |s_j|^2, \quad (1)$$

where N represents the total number of data points; M represents the number of logarithmically spaced T_2 decay times, s_j represents the amplitude of the T_2 distribution that needs to be solved; A_{ij} is the kernel matrix for the T_2 exponential decay; y_i is the measurement data; and μ is the regularization factor which was chosen to be the largest value that allows a misfit of less than $1.02\chi^2$ (MacKay et al., 1994) in our analysis (Whittall and MacKay, 1989; Hwang and Du, 2009; Levesque et al., 2010). To initialize the analysis program, 40 logarithmically spaced T_2 times in the range from 15 ms to 2s were defined as the set of basis T_2 s in equation (1). Detailed calculation steps have been described in a previous paper by Prasloski et al. (2012b). The output of the fitting program was a voxel-wise T_2 distribution and the MWF was defined as the area under the T_2 distribution between 15 ms and 40 ms relative to the total area under the entire distribution. All analysis parameters were selected using the most commonly used values in accordance with many previous studies. (Liu et al., 2019, Liu et al., 2020; Lee et al., 2018; Ljungberg et al., 2017; Dvorak et al., 2019)

2.3. Neural network model and model training

There were 7 candidate NN models with different architectures constructed using Tensorflow (Abadi et al., 2016), which is a commonly used open-source software library for deep learning. The design of all NN

models was to take the 32-echo data as input and predict MWF at the output layer, but varied in the size and the number of hidden layers (Fig. 1). All NN models were optimized by adaptive moment estimation (Adam) (Kingma and Ba, 2014) and each layer of each NN was activated by rectified linear unit (ReLU) (Nair and Hinton, 2010). The loss function was using mean squared error (MSE) and defined by the following formula:

$$Loss = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2, \quad (2)$$

where n is the total number of training examples, y_i is the NN calculation result for the i^{th} training example, and $\hat{y}_i$ is the NNLS ground truth label for the i^{th} training example. All NN models were initialized with Xavier uniform initializer (Glorot and Bengio, 2010) and the learning rate was set to 0.0005. The training data were randomly shuffled to break the data similarities of adjacent voxels and split into mini batches, each with a size of 512 training examples. Approximately 20% of total training data (600,000 voxels) were randomly selected and used as a validation set to determine the stopping criterion for training and to compare the performance of different NN architecture candidates. The training and validation set split ratio was determined empirically. Training of each NN candidate was stopped when the accuracy on the validation set did not improve further for at least 100 epochs, where improvement was defined as a decrease in MSE. The validation MSE of the 7 trained models are displayed in Table 1. The optimal NN model architecture (hidden layers: 256 × 256 × 32) was empirically chosen to achieve a balance between the validation accuracy and the risk of overfitting for deeper networks; this architecture had one of the lowest validation losses while still having a simple structure. The workflow is depicted in Fig. 1. For simplicity, we use NN_opt to denote the selected trained model of three hidden layers (256 × 256 × 32) from this point forward in this paper.

2.4. Trained model evaluation

The trained NN_opt model was applied to the testing data (2 healthy brains, 1 MS brain with segmented lesions, and 1 healthy cervical spinal cord) to produce MWF. Processing times of the trained NN_opt model and NNLS calculation of whole brain and spinal cord MWF were recorded respectively for comparison (CPU: Intel(R) Core(TM) i7-5930K @ 3.5 GHz, 32 GB RAM). For the two healthy brains, average MWFs from 6 regions of interest (ROI) (all gray matter, all white matter, corpus callosum, corticospinal tract, forceps major, forceps minor) were also extracted and compared with the NNLS ground truth. All ROIs were chosen from Johns Hopkins University (JHU) atlas (Wakana et al., 2007) and registered to GRASE-space using FSL nonlinear registration tool ‘FNIRT’. The predicted MWF values by NN_opt were compared voxel-wise with the NNLS ground truth. In particular, absolute and relative differences, correlations (R-square, t -test) between NN_opt predictions and NNLS ground truth, and mean absolute error (MAE) of all voxels were assessed. To demonstrate the robustness of the trained model, we evaluated the correlations between the NN_opt MWF errors and the actual flip angles estimated by extended phase graph simulation (Prasloski et al., 2012b). We then tested NN_opt at different SNR levels, by adding additional Rician distributed noise (SNR = 50 and 100) to the original multi-echo MWI data.

2.5. Normalized data training

Using the absolute voxel intensity for NN training raises the concern for generality because some scanner-specific factors such as amplifier gain may affect the absolute voxel intensity, especially when comparing data from different scanners, so a second model (hidden layers: 256 × 256 × 32) was trained by the 1st echo normalized data to show the effect of using 1st echo normalization. This second trained model using

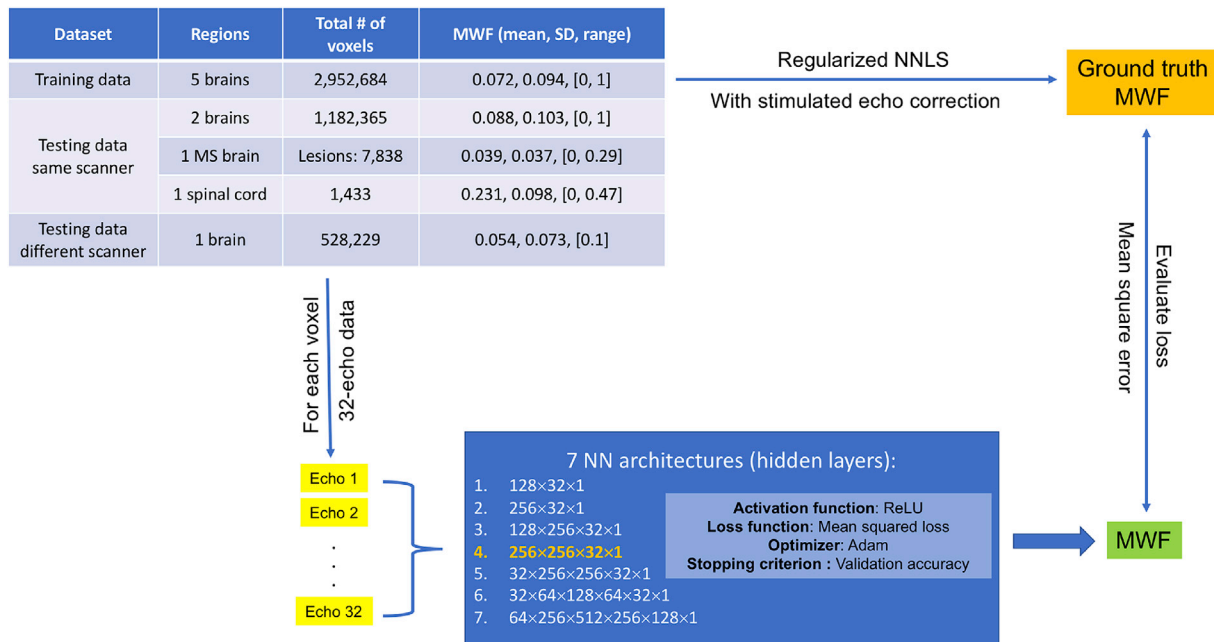


Fig. 1. Schematic of workflow. 32-echo data from 5 brains (4 healthy, 1 multiple sclerosis (MS)) were used for training. Two healthy brains, 1 MS brain with segmented lesions, and 1 spinal cord acquired by the same scanner were used as testing data. One healthy brain acquired by a different scanner were also used for testing. The ground truth was calculated by regularized non-negative least squares (NNLS) with stimulated echo correction. The neural network was constructed with fully connected layers (activation function: ReLU, optimizer: Adam, loss: mean square error).

normalized data was thus denoted as NN_norm. As an illustrative example to compare its performance with NN_opt, we only applied NN_norm to calculate the MWF maps for two healthy brains collected by Philips Achieva and Elition respectively.

2.6. Data availability

Code and trained neural network models are available upon request to the corresponding author. Data is not made publicly available due to participants confidentiality.

3. Results

3.1. MWF calculation time

The total training of the NN_opt required 46 min and 55 s at 231 epochs when the validation loss (MSE) did not decrease after 100 epochs. The trained NN_opt model was actually saved at 131 epochs (~27 min) to prevent overfitting. The processing times for the trained NN_opt model to make MWF maps were 33 s for the whole brain and 4 s for the cervical spinal cord. In contrast, NNLS analysis took 1 h and 33 min for the whole brain and 15 min for the cervical spinal cord. The data processing speed of the trained NN_opt model was more than 150-fold faster than that of the NNLS algorithm.

Table 1

Validation loss of 7 NN models. 7 candidate NN models with different hidden layers were constructed and trained respectively. The training stopped when the validation loss (mean squared error) did not decrease for 100 epochs.

Model	Validation loss
128 × 32 × 1	8.35×10^{-5}
256 × 32 × 1	7.26×10^{-5}
128 × 256 × 32 × 1	5.04×10^{-5}
256 × 256 × 32 × 1	4.53×10^{-5}
32 × 256 × 256 × 32 × 1	5.39×10^{-5}
32 × 64 × 128 × 64 × 32 × 1	4.29×10^{-5}
64 × 256 × 512 × 256 × 128 × 1	4.16×10^{-5}

3.2. Healthy brains

Fig. 2 shows excellent visual correspondence between MWF maps from the NN_opt model, NN_norm model and NNLS analysis. No visual differences were observed for all slices in the MWF maps for both trained NN_opt and NN_norm models. Slight regional biases were only noticeable in the frontal regions in the error maps which were viewed with rescaling the colormap. It's also observed that the error of NN_opt (MAE = 0.004) was lower than that of NN_norm (MAE = 0.008). The NN_opt model MWF predictions and the NNLS ground truth from 6 ROIs of the two testing healthy brains are summarized in Table 1. Using one of the healthy brains as an illustrative example, voxel-wise NN_opt MWF correlated almost perfectly with NNLS MWF in the 6 different ROIs (average $R^2 = 0.994$, $p < 0.0001$, MAE = 0.003, Fig. 3).

3.3. MS lesions

The absolute and relative differences in the average MWF of the lesions (trained NN_opt: (0.0396 ± 0.0366) ; NNLS ground truth: (0.0393 ± 0.0365)) were 0.0003 and 0.76% respectively, and a strong voxel-wise correlation between the two calculation methods were observed ($R^2 = 0.987$, $p < 0.001$, MAE = 0.003, Fig. 4).

3.4. Spinal cord

Average MWF (0.2319 ± 0.0981) for the cervical spinal cord C2–C3 segment calculated by the trained NN_opt were very similar to NNLS ground truth (0.2306 ± 0.0975). The absolute and relative differences in whole cord MWF average were 0.0013 and 0.56% respectively. Excellent voxel-wise correlation between the two MWF calculation methods were found ($R^2 = 0.988$, $p < 0.001$, MAE = 0.0076, Fig. 5).

3.5. Data from a different scanner

In this illustrative example, the MWF maps of one healthy brain collected by a different scanner (Philips Elition) were carried out by NN_opt, NN_norm and NNLS ground truth respectively, and presented in

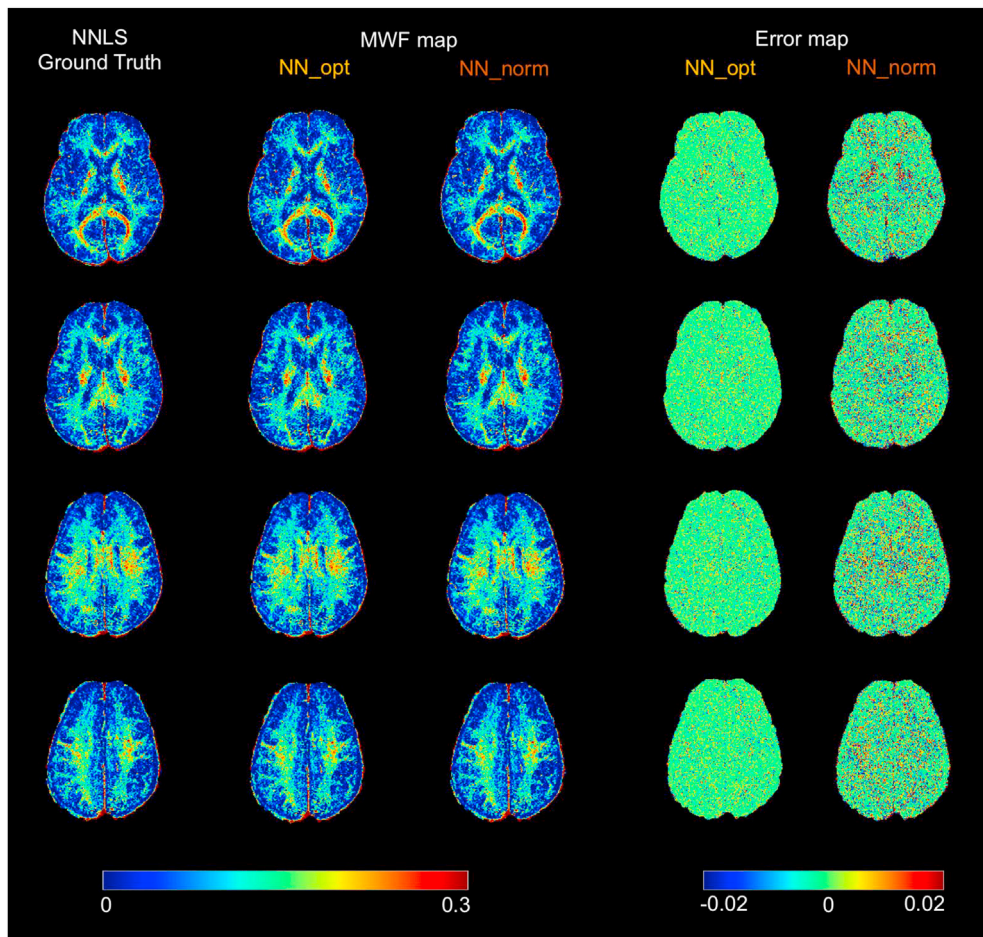


Fig. 2. Visual comparison of brain MWF maps. The brain MWF maps carried out by the neural network models (NN_opt and NN_norm) and the NNLS ground truth, as well as the error map between the two are shown side by side for 4 representative axial slices from healthy brain #1. Note that NN_opt was trained by non-normalized data while NN_norm was trained by normalized data.

Fig. 6, alongside with the error maps. Although the voxel-wise MAE were increased to 0.0049 for NN_opt and 0.0108 for NN_norm and errors occurred primarily in white matter, both models were still capable of producing visually very similar MWF maps as NNLS, with only subtle differences visible if examined closely.

3.6. Robustness of the trained model

Using one brain (collected by a Philips Achieva system) as an example, we plotted the prediction error of MWF made by the NN_opt against the flip angle in each brain voxel and found no apparent correlation ($R^2 < 0.001$, $p < 0.001$) between the two (Fig. 7). The same behaviour was observed with the other testing examples, including data collected using a different scanner (Philips Elition). Injecting noise at different SNR levels (SNR = 50 and 100) into the original brain data, we found NN_opt was able to produce virtually the same MWF maps as NNLS, even at low SNR levels (SNR = 50) (Fig. 8).

4. Discussion

4.1. Prediction accuracy

For all the regions that were tested, including healthy brains (from both scanners), MS lesions, and cervical spinal cord, MWF maps calculated by the trained NN_opt model exhibit no qualitative differences compared with the NNLS ground truth MWF maps (Figs. 2, 4–6). Quantitatively, almost all voxels within each ROI exhibited excellent

agreement (all $R^2 > 0.98$, $p < 0.001$) between NN_opt and NNLS (Figs. 3–5). For the healthy brains, absolute differences of mean MWF can only be found after four decimal places and the relative differences were $< 0.25\%$ for white matter regions (Table 2), much smaller than the actual uncertainty of NNLS in estimating the true MWF. For the MS lesions, where low MWF values are expected due to demyelination and greater tissue heterogeneity exists compared to unaffected white matter (Keegan and Noseworthy, 2002; Laule et al., 2004), the trained NN_opt model still performed accurate MWF calculations with a relative difference of less than 1%. For the cervical spinal cord, we observed a slightly higher absolute difference (approximately 0.001) of NN_opt MWF compared to NNLS ground truth, but the relative difference was low ($< 1\%$). Similarly, in our illustrative example where brain data were collected by a different scanner, the trained NN_opt could still produce accurate MWF maps with slightly increased errors. Although training data only included brain data from one scanner, the trained NN_opt was still able to overcome the challenge of these inherent differences in both the spinal cord and the different scanner data, and performed well in calculating the MWF without any data normalization. To summarize, the extremely low error of the trained NN_opt model demonstrates that it is capable of producing virtually the same MWF maps as NNLS for the brain, pathological lesions, spinal cord, and data collected from a different scanner.

4.2. NN_opt vs. NN_norm

It is observed that the MWF values calculated by the NN_norm model have larger errors than that of the NN_opt model (Figs. 2 and 6), raising a

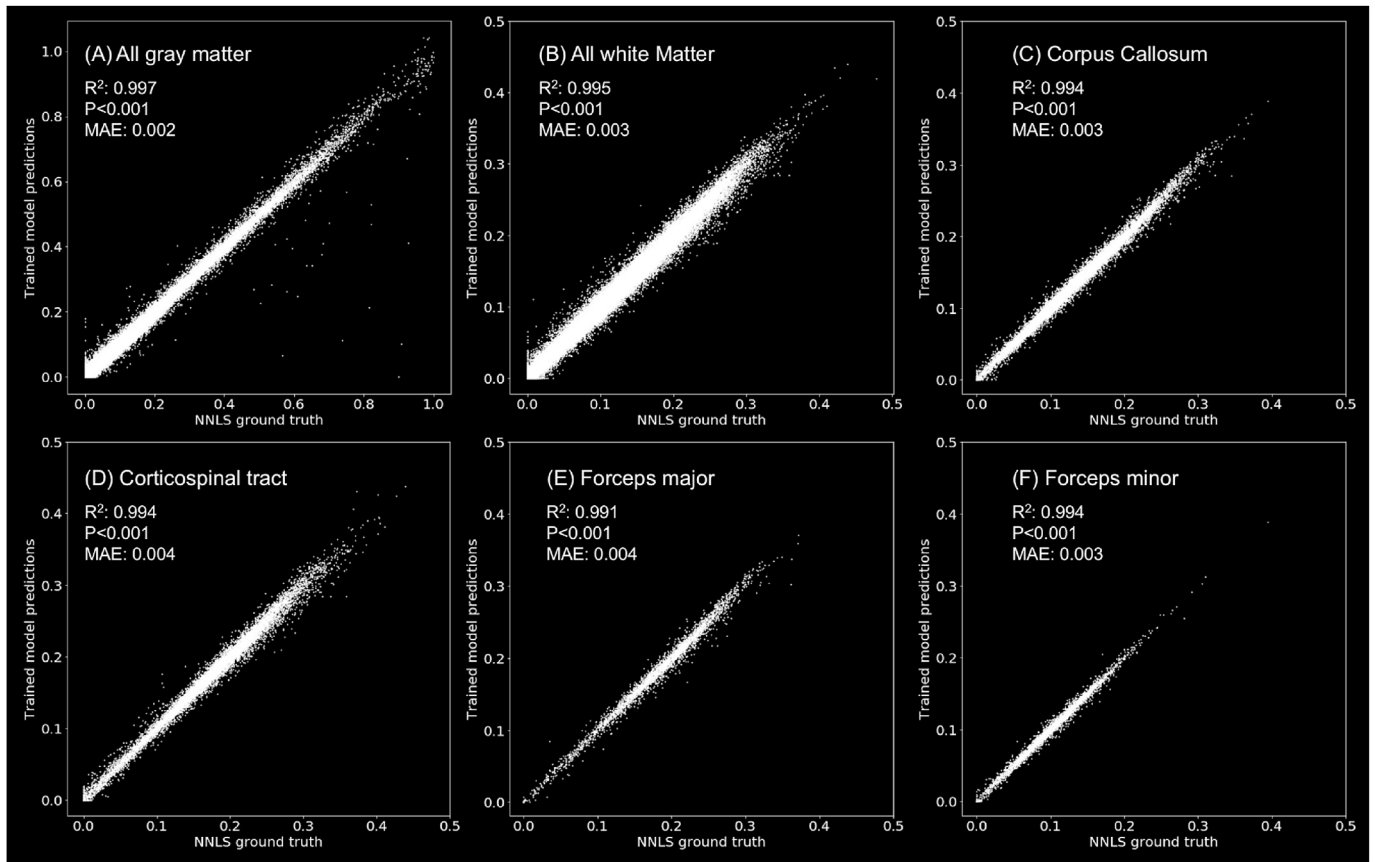


Fig. 3. Voxel-wise comparison of MWF within six different brain ROIs. ROIs include all gray matter, all white matter, corpus callosum, corticospinal tract, forceps major, and forceps minor from healthy brain #2. R-square, p values from paired *t*-test, and mean absolute errors (MAE) were evaluated.

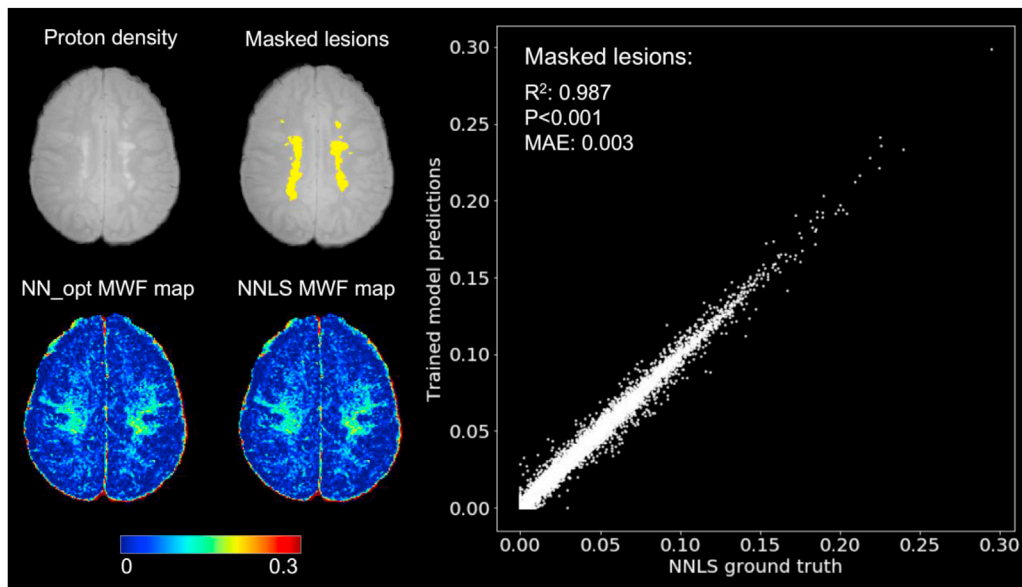


Fig. 4. MWF comparison for lesions. The lesion mask was first manually drawn on the proton density image of the MS brain and then registered to the GRASE space. MWF maps carried out by the trained neural network model (NN_opt) and the NNLS ground truth are visually the same. Highly significant voxel-wise MWF correlation between NN_opt and NNLS is shown for the lesions.

concern about the 1st echo normalization method used in NN_norm. Normalization to a proton density weighted image may be an alternative, assuming the within-subject image registration can be done successfully. Note that the first echo image acquired at 10 ms in our study cannot be

regarded as an ideal proton density weighted image because the TR used in our sequence was 1 s, which is not long enough to fully eliminate the T_1 weighting. Moreover, the excellent performance of NN_opt on data collected from a different scanner with a different coil (Fig. 6)

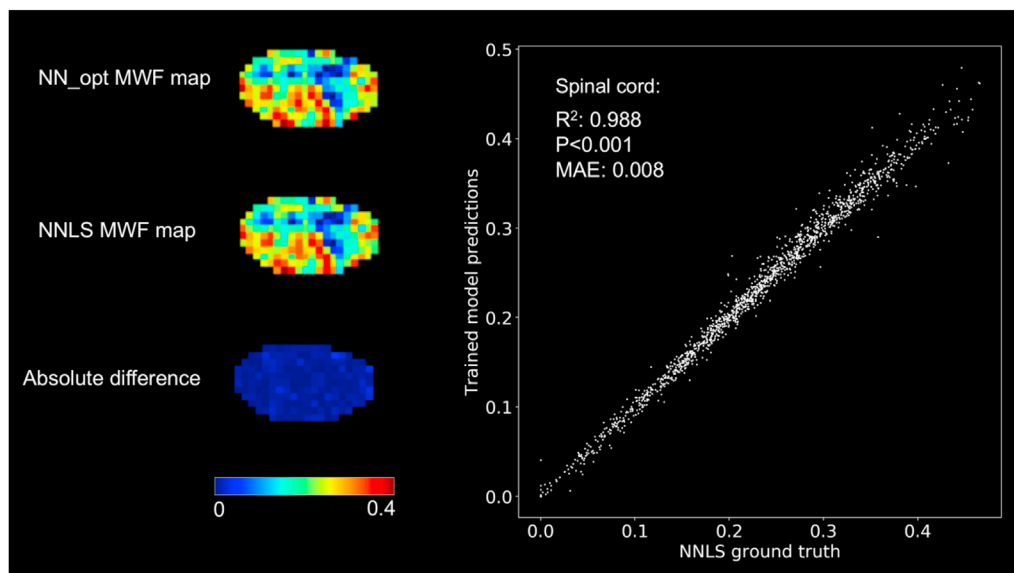


Fig. 5. Comparison of spinal cord MWF. The MWF maps of a healthy spinal cord were carried out by the neural network model (NN_opt) and the NNLS ground truth. No visual differences or regional biases were found. Excellent voxel-wise MWF correlation between NN_opt and NNLS was observed in the spinal cord.

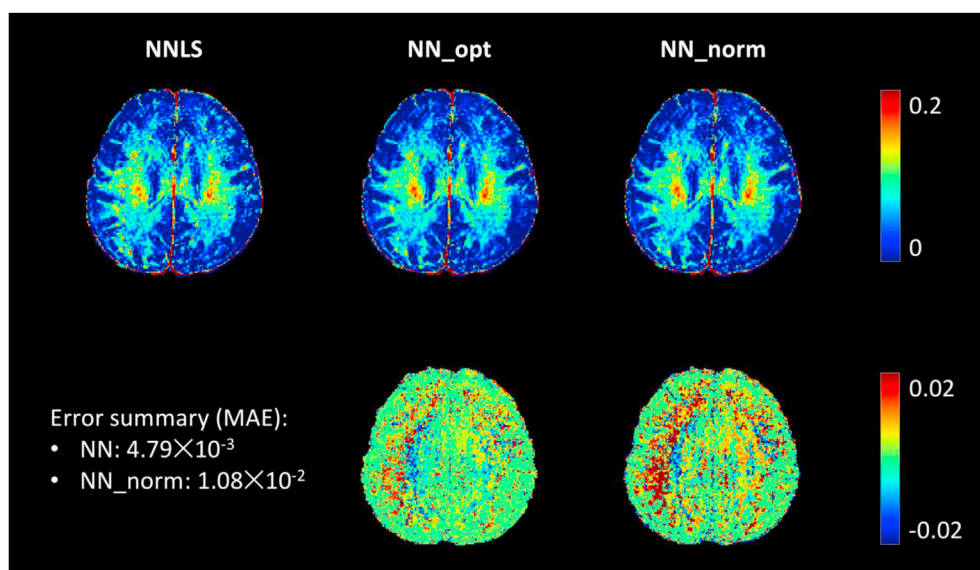


Fig. 6. Analysis of brain data acquired by a different scanner. The brain MWF maps carried out by the neural network models (NN_opt and NN_norm) and the NNLS ground truth, as well as the error maps are shown side by side for one representative axial slice from a healthy brain data collected by a different scanner (Philips Elition). Note that NN_opt was trained by non-normalized data while NN_norm was trained by normalized data.

demonstrates the robustness of the trained NN_opt. However, slightly enlarged MAE (4.79×10^{-3}) of the NN_opt model on the different scanner data indicates the trained NN_opt may have some biases towards different scanners, but not to the extent of corrupting accurate MWF predictions. This finding also raises the question to what extent the trained NN_opt can tolerate variations of the absolute intensity in the data collected from different scanners. It is unlikely that NN_opt would be completely reliable on all systems due to the somewhat random effects of receiver gains on overall signal intensity. One may propose to multiply the different scanner data by a constant scaling factor to match the intensity level of the training data, and then apply the NN_opt for a more accurate MWF calculation. Our post-hoc consideration of such an approach found such data scaling did not work well and generated unreasonable MWF maps, revealing that the data scaling step may likely worsen the performance of NN_opt. From this perspective, NN_norm has

an intrinsic advantage for being truly scale invariant and is thus safer to use when it comes to data from a different scanner. Collectively, when a different scanner is used, fine-tuning or further training of the existing NN_opt and NN_norm models by including training data from that particular scanner is certainly encouraged. Our illustrative example demonstrates the feasibility of this approach, but a multi-vendor study with more participants is warranted to explore this further.

4.3. Trained NN_opt sensitivity to flip angle and SNR

Although the NNLS ground truth was produced with B1 inhomogeneity correction, how resilient the trained NN_opt is to the actual flip angle was a reasonable concern. We addressed this concern by showing that the prediction accuracy of the trained NN_opt has no dependency on the actual flip angle (Fig. 7). Another possible concern

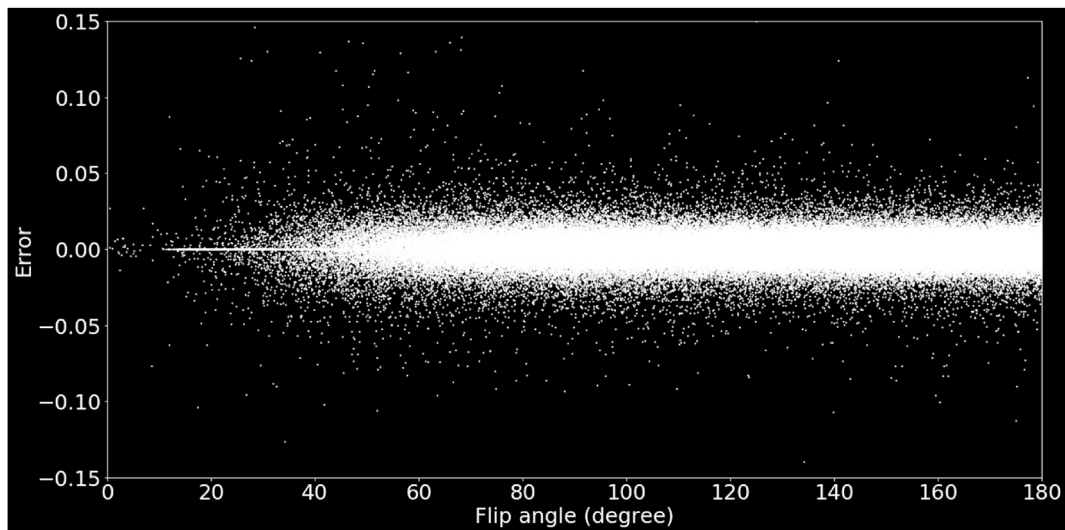


Fig. 7. Neural network prediction error vs. flip angle. The actual refocusing flip angle deviates from the prescribed 180° due to B1 inhomogeneity. The voxel-wise flip angle was estimated using extended phase graph algorithm. The neural network (NN_opt) prediction error is plotted against the actual flip angle.

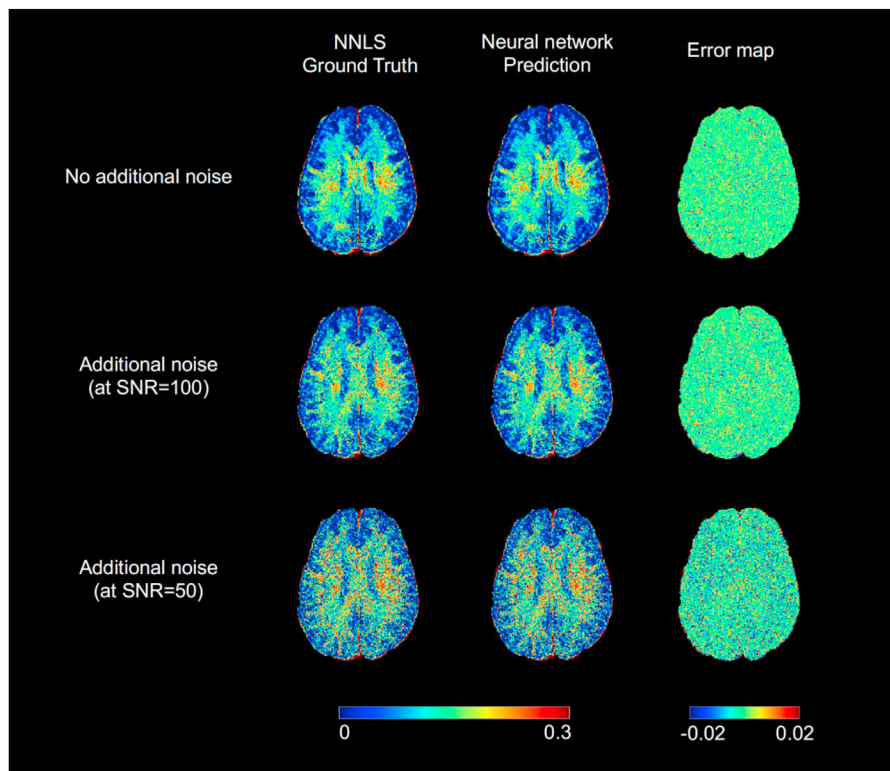


Fig. 8. Neural network prediction at different SNR levels. Additional rician noise is injected at SNR = 100 and SNR = 50. Neural network (NN_opt) predictions are compared with NNLS ground truth. MWF and its error maps of one representative axial slice are shown.

was the robustness of the NN_opt to decreased SNR. Thus, we tested the trained model by adding additional noise to the original data. It was observed that the NN_opt model is still capable of producing visually the same MWF maps as NNLS even when SNR is poor (additional noise at SNR = 50) and the NNLS ground truth algorithm starts to fail (Fig. 8). Overall, interrogations on the flip angle and decreased SNR demonstrate the proposed NN_opt model is a robust representation of the NNLS algorithm with stimulated echo correction and regularization.

4.4. Implementation: neural network vs. NNLS

The NNLS algorithm has been extensively applied in many fields and the code for NNLS is mostly open source accessible. However, in most scenarios with MWI, simply applying the exact open source code of NNLS will likely produce unsatisfactory results. In recent advancement of MWI analysis, the NNLS algorithm has been incorporated with stimulated echo correction (Prasloski et al., 2012b) for more accurate MWF estimations, and regularization is usually applied (Du et al., 2007; Whittall et al.,

Table 2

Summary of results for the two testing healthy brains. Absolute and relative differences were evaluated.

Healthy brain 1	# of voxels	NNLS ground truth	Absolute MWF differences: NN_opt vs. NNLS	Relative MWF differences: NN_opt vs. NNLS
All gray matter	299,298	0.07 ± 0.09	0.00009	0.13%
All white matter	168,675	0.11 ± 0.06	0.00004	0.04%
Corpus callosum	19,778	0.13 ± 0.06	0.00005	0.04%
Corticospinal tract	8,898	0.17 ± 0.07	0.00016	0.09%
Forceps major	3,245	0.16 ± 0.06	0.00012	0.08%
Forceps minor	3,285	0.11 ± 0.06	0.00028	0.24%

Healthy brain 2	# of voxels	NNLS ground truth	Absolute MWF differences: NN_opt vs. NNLS	Relative MWF differences: NN_opt vs. NNLS
All gray matter	337,140	0.06 ± 0.09	0.00008	0.14%
All white matter	199,637	0.10 ± 0.07	0.00007	0.07%
Corpus callosum	18,709	0.13 ± 0.06	<0.00001	<0.01%
Corticospinal tract	11,851	0.16 ± 0.08	0.00015	0.09%
Forceps major	3,356	0.18 ± 0.06	0.00040	0.22%
Forceps minor	3,414	0.10 ± 0.05	0.00001	0.01%

$p < 0.000001$ for all NNLS vs NN_opt comparisons.

1997; Yoo and Tam, 2013) for smoother appearing maps. These integrations make the modern NNLS analysis mathematically complex and computationally expensive. Customization of open source code by users is almost always required to include stimulated echo correction and regularization, making the modern NNLS implementation nontrivial. Therefore, in addition to its lengthy data processing time, the implementation complexity also prevents NNLS analysis from being widely used. To overcome this hurdle, our proposed NN_opt was trained with regularized NNLS ground truth labels that have been corrected for stimulated echoes. As a result, the trained NN_opt model is inherently embedded with features of regularization and stimulated echo correction, thus no further customization is needed. Besides, from a mathematical point of view, what the trained NN model does to the imaging data is essentially a few simple steps of multiplication and addition. Compared with the optimization process of conventional fitting algorithms, it is computationally inexpensive and can be easily implemented on various platforms (matlab, python etc.) with common hardware compatibility. Finally, the trained NN_opt model is super lightweight (file size for our NN_opt model: 80 KB) so it can be easily stored and distributed either online or locally.

4.5. Limitations of current model

A few limitations need to be considered. (1) The trained NN_opt model inherits all of the NNLS algorithm's defects. For example, the current NNLS analysis is susceptible to noise so high-quality data is required in order to obtain a reliable MWF estimation. The trained NN_opt model suffers the same problem since we used NNLS results as the ground truth for training. (2) For this proof of concept study, we only trained the NN_opt with one set of initialization parameters for NNLS analysis (number of T_2 s = 40, T_2 range = [15, 2000] ms, and MWF window = [15, 40] ms), on data collected using one version of the GRASE sequence. Although NNLS initialization parameters were set to the most

commonly used values, there are still occasions, such as applying MWI to spinal cord (MacMillan et al., 2011) and post-mortem tissue samples (Laule et al., 2016), when adjustments to these initialization parameters may be needed. Nevertheless, this problem can be fixed by training multiple NN models using different sets of initialization parameters since the one-time training time is acceptable. Training of new NN models can be accelerated by transfer learning, where our existing NN_opt and NN_norm models are used as the starting point for training the new model, due to the similarities between these two tasks. The transfer learning approach also applies when data acquisition sequence parameters are modified. (3) NNLS analysis is able to output a T_2 distribution instead of just MWF. Many other valuable quantities such as total water content (Laule et al., 2004) or geometric mean T_2 of IE water (MacMillan et al., 2011) can also be extracted from the voxel-wise T_2 distribution. An important next step will be constructing and training a multi-output NN model that can make calculations of all of these other important quantities.

4.6. Other quantitative MRI techniques

Similar to MWI data analysis, many other quantitative MRI (qMRI) techniques, such as neurite orientation dispersion and density imaging (NODDI) (Zhang et al., 2012) and multicomponent driven equilibrium single pulse observation of T_1 and T_2 (mcDESPOT) (Deoni et al., 2008), are also mathematically complex and computationally expensive for their image post-analysis. Besides the reliability issues of these qMRI techniques for use in single individuals, lengthy data processing time is another common problem. In our opinion, the methodology of using a NN to create a fast and compact representation of a complex mathematical function can also be easily adopted by other qMRI data analysis to significantly shorten the image post-processing time.

4.7. Comparison to kernel regression

Kernel regression (KR) (Saunders et al., 1998) is a classical machine learning method that also has the ability to map any qMRI data to the desired output, in theory similar to NN. Due to the non-iterative nature of the training process, a conventional KR model can be trained in seconds, which seems to be a huge advantage over the NN method. Another advantage of KR is that it is mathematically explainable, unlike NN which is a black box operation. However, the conventional KR method is only suitable for a small or moderate dataset and has problems with scaling up. A large dataset like the one used in this study can cause memory and storage problems, leading to an unrealistic training time (Ali and Recht, 2008). To overcome this hurdle, modern kernel approximation techniques such as random fourier (Ali and Recht, 2008) can be applied to effectively accelerate KR training. Based on KR method equipped with random fourier kernel approximation, Nataraj et al. proposed a framework, namely parameters estimation via regression with kernels (PERK), to calculate qMRI parameters such as steady state MWF. (Nataraj et al., 2018b, <https://github.com/gopal-nataraj/perk>) The reported KR model training time for calculating steady state MWF was 3.6 s on 100,000 training examples with a dimensionality of 4. Assuming the same KR model were used, and scalability were linear (at best in theory, usually worse in practice), then the estimated training time on our GRASE data, which has 3 million training examples with a dimensionality of 32, would be at least 14.4 min (linearly scaled up by 240 times). However, the accuracy of MWF predictions cannot be compared here because the KR model was not trained on NNLS labels. Nevertheless, NN is able to intrinsically handle massive datasets with high dimensionalities. It can be constructed, trained and distributed by commonly used platforms such as Tensorflow, which only requires a minimum level of coding skills to use. On the other hand, how much training data is adequate for training NN and KR models remains an open question. A comprehensive comparison between NN and KR methods is beyond the scope of this study and we would leave the choice of method to the readers.

5. Conclusion

We have demonstrated that MWI data processing time can be dramatically reduced from hours to less than 1 min (over $150 \times$ acceleration) without degrading the analysis results by using a trained NN model, which has been tested for healthy brain and spinal cord, pathological lesions, as well as data acquired from a different scanner. Compared to conventional NNLS analysis incorporated with stimulated echo correction and regularization, our trained NN models (NN_opt and NN_norm) are considerably easier to implement. Finally, the proposed method of using a NN model to shorten the processing time can be used as a universal methodology for many other qMRI data analyses.

Author contributions

Hanwen Liu, study concept and design, acquisition of data, analysis and interpretation of data, drafting manuscript; Qing-San Xiang, study concept and design, data interpretation, critical revision of manuscript for intellectual content; Roger Tam, study concept and design, data interpretation, critical revision of manuscript for intellectual content; Adam V. Dvorak, acquisition of data, critical revision of manuscript for intellectual content; Alex L. MacKay, study concept and design, critical revision of manuscript for intellectual content; Shannon H. Kolind, acquisition of data, critical revision of manuscript for intellectual content; Anthony Traboulsee, acquisition of data, critical revision of manuscript for intellectual content; Irene M. Vavasour, acquisition of data, critical revision of manuscript for intellectual content; David K.B. Li, study concept and design, critical revision of manuscript for intellectual content; John K. Kramer, study concept and design, acquisition of data, critical revision of manuscript for intellectual content; Cornelia Laule, study concept and design, data interpretation, drafting manuscript, critical revision of manuscript for intellectual content.

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